THE UNIT OF THE ADJUNCTION COMMUTES WITH BASE CHANGE THROUGH THE COMPARISON TRANSFORMATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6K (sources/stacks-duality/pinned-original-source.tex line 1956).

Allowed standard prerequisites (editorial).

- The base change diagram and the comparison transformations alpha and beta. Same chapter, equation labelled equation-base-change-code, diagram labelled equation-base-change, equation labelled equation-base-change-map and Lemma labelled lemma-flat-precompose-pus, cited at proof lines 1978-1992. Fix the two transformations whose agreement is asserted.
- The ordinary base change remark for pushforward. Cohomology, Remark labelled cohomology-remark-base-change, invoked in the statement. Supplies beta and its isomorphism property.
- The dual trace map lemma. Same chapter, Lemma labelled lemma-trace-map-and-base-change, cited at proof lines 1978-1980. Provides the dual argument being imitated.
- Coherence for compositions of functors and the transformation between adjoints. Categories, Lemma labelled categories-lemma-properties-2-cat-cats and Definition labelled categories-definition-horizontal-composition, cited at proof lines 2014-2018, and Categories, Lemma labelled categories-lemma-transformation-between-functors-and-adjoints, cited at proof lines 2020-2024. Give the commutativity of the squares and of the pentagon in the proof diagram.

Source reference. Original equation 0A9J: sources/stacks-duality/pinned-original-source.tex, source lines 927–927. Same-chapter pointer into the author environment equation at source line 927; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 08HY: sources/stacks-chow/tags.txt, source lines 10796–10796. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0AA6: sources/stacks-duality/pinned-original-source.tex, source lines 1192–1192. Same-chapter pointer into the author environment equation at source line 1192; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6J: sources/stacks-duality/pinned-original-source.tex, source lines 1867–1867. Same-chapter pointer into the author environment lemma at source line 1867; invoked at proof line 1979 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9K: sources/stacks-duality/pinned-original-source.tex, source lines 957–957. Same-chapter pointer into the author environment lemma at source line 957; invoked at proof line 1986 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 003F: sources/stacks-chow/tags.txt, source lines 132–132. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2015 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 003E: sources/stacks-chow/tags.txt, source lines 131–131. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2017 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B65: sources/stacks-chow/tags.txt, source lines 14058–14058. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2020 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Suppose we have a diagram (0A9J) where f and g are tor independent. Then the maps $1 \star \eta_f : L(g')^* \to L(g')^* \circ a \circ Rf_*$ and $\eta_{f'} \star 1 : L(g')^* \to a' \circ Rf'_* \circ L(g')^*$ agree via the base change maps $\beta : Lg^* \circ Rf_* \to Rf'_* \circ L(g')^*$ (Cohomology, Remark 08HY) and $\alpha : L(g')^* \circ a \to a' \circ Lg^*$ (0AA6). More precisely, the diagram*

$$\begin{array}{ccc} L(g')^* & \xrightarrow{1 \star \eta_f} & L(g')^* \circ a \circ Rf_* \\ {\scriptstyle \eta_{f'} \star 1}\downarrow & & \downarrow{\scriptstyle \alpha} \\ a' \circ Rf'_* \circ L(g')^* & \xleftarrow{\beta} & a' \circ Lg^* \circ Rf_* \end{array}$$

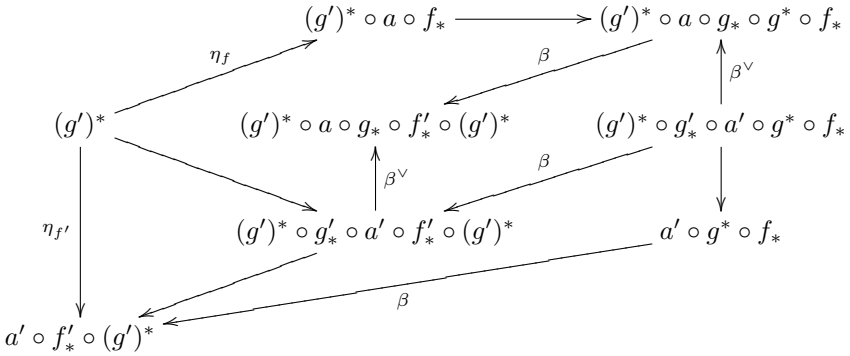
of transformations of functors commutes.

Proof. This proof is dual to the proof of Lemma 0B6J. In this proof we write f_* for Rf_* and g^* for Lg^* and we drop $\star$ products with identities as one can figure out which ones to add as long as the source and target of the transformation is known. Recall that $\beta : g^* \circ f_* \to f'_* \circ (g')^*$ is an isomorphism and that α is defined using the isomorphism $\beta^\vee : g'_* \circ a' \to a \circ g_*$ which is the adjoint of β, see Lemma 0A9K and

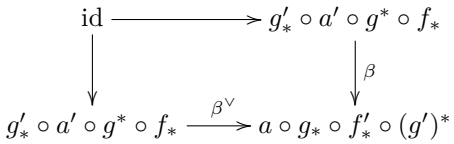
its proof. First we note that the left vertical arrow of the diagram in the lemma is equal to the composition

$$(g')^* \to (g')^* \circ g'_* \circ (g')^* \to (g')^* \circ g'_* \circ a' \circ f'_* \circ (g')^* \to a' \circ f'_* \circ (g')^*$$

where the first arrow is the unit for $((g')^*, g'_*)$, the second arrow is $\eta_{f'}$, and the third arrow is the counit for $((g')^*, g'_*)$. This is a simple consequence of the fact that the composition $(g')^* \to (g')^* \circ (g')_* \circ (g')^* \to (g')^*$ of unit and counit is the identity. Consider the diagram



In this diagram the two squares commute Categories, Lemma 003F or more simply the discussion preceding Categories, Definition 003E. The triangle commutes by the discussion above. By the dual of Categories, Lemma 0B65 the square



commutes which implies the pentagon in the big diagram commutes. Since β and $\beta^\vee$ are isomorphisms, and since going on the outside of the big diagram equals $\beta \circ \alpha \circ \eta_f$ by definition this proves the lemma. $\square$